CLASS 10 · CBSE · NCERT MATHEMATICS · SHORT NOTES

Chapter 12: Surface Areas and Volumes

Topic-wise short notes with solved examples

Real objects are often two or more basic solids joined together. This chapter shows how to find the surface area and the volume of such combinations.

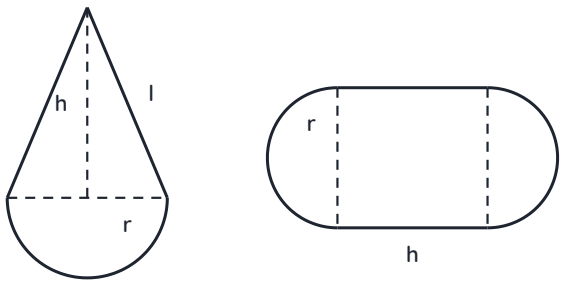
1 Formulas for the basic solids

Solid	Curved / lateral surface area	Total surface area	Volume
Cuboid (l, b, h)	$2h(l+b)$	$2(lb+bh+hl)$	lbh
Cube (side a)	$4a^2$	$6a^2$	a^3
Cylinder (r, h)	$2\pi rh$	$2\pi r(r+h)$	$\pi r^2 h$
Cone (r, h, l)	πrl	$\pi r(l+r)$	$\frac{1}{3}\pi r^2 h$
Sphere (r)	$4\pi r^2$	$4\pi r^2$	$\frac{4}{3}\pi r^3$
Hemisphere (r)	$2\pi r^2$	$3\pi r^2$	$\frac{2}{3}\pi r^3$

- For a cone, the **slant height** is $l = \sqrt{r^2 + h^2}$.
- Use $\pi = \frac{22}{7}$ unless the question gives another value.
- 1 litre $= 1000$ cm³ and 1 m³ $= 1000$ litres.

2 Surface area of a combination of solids

- The total surface area of a combined solid is the sum of the **visible** surfaces only.
- The faces along which the solids are joined are hidden, so they are **not** counted.
- Usually this means adding curved surface areas, plus any flat face that remains exposed.



Left: a cone on a hemisphere. Right: a cylinder with hemispherical ends.

Combination	Surface area
Cone on a hemisphere (same radius)	$\pi rl + 2\pi r^2$
Cylinder with a hemisphere at each end	$2\pi rh + 4\pi r^2$
Cylinder with a cone on top (tent)	$2\pi rh + \pi rl$ (plus πr^2 if the base is included)
Cube with a hemisphere placed on one face	$6a^2 + \pi r^2$

Example 2.1 A toy is a cone mounted on a hemisphere of the same radius 7 cm. The total height of the toy is 31 cm. Find its total surface area.

SOLUTION

Height of the cone $= 31 - 7 = 24$ cm, so $l = \sqrt{7^2 + 24^2} = \sqrt{625} = 25$ cm

CSA of cone $= \pi rl = \frac{22}{7} \times 7 \times 25 = 550$ cm²

CSA of hemisphere $= 2\pi r^2 = 2 \times \frac{22}{7} \times 49 = 308$ cm²

Total surface area $= 550 + 308 = 858$ **cm²**

Example 2.2 A capsule is a cylinder with a hemisphere at each end. Its total length is 20 cm and its diameter is 7 cm. Find its surface area.

SOLUTION

$r = 3.5$ cm. Height of the cylinder $= 20 - 2 \times 3.5 = 13$ cm

Surface area $= 2\pi rh + 4\pi r^2 = 2\pi r(h + 2r) = 2 \times \frac{22}{7} \times 3.5 \times (13 + 7)$

Surface area $= 22 \times 20 = 440$ **cm²**

Example 2.3 A hemispherical depression of diameter 7 cm is cut out from one face of a wooden cube of side 7 cm. Find the surface area of the remaining solid.

SOLUTION

Surface area = TSA of cube − area of the circle removed + CSA of the hemisphere

$= 6a^2 - \pi r^2 + 2\pi r^2 = 6a^2 + \pi r^2$

$= 6 \times 49 + \frac{22}{7} \times 3.5 \times 3.5 = 294 + 38.5$

Surface area $= 332.5$ **cm²**

Example 2.4 A tent is a cylinder of radius 7 m and height 4 m with a cone of slant height 13 m on top. Find the area of canvas needed.

SOLUTION

Canvas = CSA of cylinder + CSA of cone $= 2\pi rh + \pi rl = \pi r(2h + l)$

$= \frac{22}{7} \times 7 \times (8 + 13) = 22 \times 21$

Area of canvas $= 462$ **m²**

REMEMBER
Before adding, ask of each surface: "Can it be seen or touched from outside?" If not, leave it out.

3 Volume of a combination of solids

- The volume of a combined solid is the **sum** of the volumes of its parts. Nothing is hidden, so nothing is left out.
- If a part is scooped out or drilled out, **subtract** its volume.

Example 3.1 Find the volume of the toy in the first example of topic 2 (a cone of height 24 cm on a hemisphere, both of radius 7 cm).

SOLUTION

Volume of cone $= \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times 49 \times 24 = 1232$ cm³

Volume of hemisphere $= \frac{2}{3}\pi r^3 = \frac{2}{3} \times \frac{22}{7} \times 343 = \frac{2156}{3} \approx 718.67$ cm³

Volume $\approx 1232 + 718.67 = 1950.67$ **cm³**

Example 3.2 Find the volume of the capsule in the second example of topic 2 (radius 3.5 cm, cylinder height 13 cm).

SOLUTION

Volume of cylinder $= \pi r^2 h = \frac{22}{7} \times 3.5 \times 3.5 \times 13 = 500.5$ cm³

Volume of the two hemispheres $= \frac{4}{3}\pi r^3 = \frac{4}{3} \times \frac{22}{7} \times 42.875 \approx 179.67$ cm³

Volume $\approx 500.5 + 179.67 = 680.17$ **cm³**

Example 3.3 A conical cavity of the same height and the same base is hollowed out of a solid cylinder of height 12 cm and radius 5 cm. Find the volume and the total surface area of the remaining solid. Leave the answers in terms of π.

SOLUTION

Volume $= \pi r^2 h - \frac{1}{3}\pi r^2 h = \frac{2}{3}\pi r^2 h = \frac{2}{3} \times \pi \times 25 \times 12 = 200\pi$ cm³

Slant height of the cone: $l = \sqrt{5^2 + 12^2} = 13$ cm

TSA $=$ CSA of cylinder $+$ area of the base $+$ CSA of the cone $= 2\pi rh + \pi r^2 + \pi rl = \pi(120 + 25 + 65)$

Volume $= 200\pi$ **cm³; total surface area** $= 210\pi$ **cm²**

4 Working with the dimensions

- Find the height of each part from the total height: for a cone on a hemisphere, cone height $=$ total height $- r$.
- For a cylinder with hemispherical ends, cylinder height $=$ total length $- 2r$.
- The greatest diameter of a hemisphere on a cube face, or of a sphere inside a cube, equals the side of the cube.
- Volume of water displaced by a solid that is fully immersed equals the volume of the solid.

5 Common mistakes

- Adding **total** surface areas of the parts. Only the exposed surfaces count.
- Using the vertical height h in πrl. The cone's curved surface needs the slant height l.
- Using the diameter where the radius is needed.
- Forgetting to subtract r or $2r$ from the total height.
- Mixing units, such as cm with m, or giving a volume in cm².

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